Project

Analysis and Insights of LAX Flight Delays 2015

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1.1 Business goals and objectives

Historical domestic flight data for flights departing from Los Angeles International (LAX) airport in 2015 was provided to investigate the reasons for flight delays. The overall goal of the data analysis is to advise a company specialising in last-minute weekend flights in the USA on how to improve customer experience, enhance service quality, reduce costs and minimise landing and departure bottlenecks. Specifically, the analysis will investigate the causes of delays and the following influences:

- destination airport,
- destination state,
- airline,
- number of departing flights,
- time of day,
- day of the week, and
- month of the year.

1.2 Data provided

The flight data was provided as an .xls file with variable names, descriptions, types and classes in Appendix 1, Table 2. Key variables related to delay causes include air-system delay, security delay, airline delay, late-aircraft delay and weather delay with definitions in Appendix 1, Table 2 (Bureau of Transportation Statistics, n.d.).

Additionally, two files, airlines.csv and airports.csv contain International Air Transport Association's (IATA) airline and airport codes (Appendix 1, Table 3 and Table 4). Since the airports file only lists state abbreviations, a fourth file from GitHub was obtained, providing state names and abbreviations (Ong, n.d.) (Appendix 1, Table 5).

1.3 Assess quality of the data

Excel and python were utilised to determine missing values, inconsistencies and anomalies in the data. Refer to Table 1 for missing values and anomalies in the flight data and the cleaning approach (See also Section 3.2 in Appendix).

Table 1 Missing values, anomalies and cleaning approach for flightdata

Variable	Missing values	Anomalies found	Cleaning approach
YEAR	0	There are 9 entries entered incorrectly with either 2035 or 2025	Corrected the Year by viewing the SCHEDULED DEPARTURE_TIME value in the same entry which also includes the date.
MONTH	0	Values should range from 1=January to 12=December, However found three months with incorrect values >12 No October dates were provided.	Viewed the SCHEDULED DEPARTURE_TIME value in the same entry which also includes the date and corrected not only the MONTH VALUE but also the DAY as this was entered incorrectly also. It seems Month should have been day and vice versa.
DAY	0	Values should range from Min=0 to Max=31 Three values were entered incorrectly which were found due to action above.	Corrected as above
DAY_OF_WEEK	3	Values should range from 1=Monday, 2=Tuesday to 7=Sunday In sorting found two values>10, 9 values >12 4 values=WEDS	Created a column called Actual_Date and used the formula =DATE(A2, B2, C2), where A=YEAR, B=MONTH, C=DAY to calculate actual date. Created a new column called ACTUAL_DAYOF WEEK and calculated day of the week by using the formula =TEXT(D2,"dddd") Replaced the values in the DAY_OF_WEEK with the correct numerical values for the day of the week.
IATA_CODE_AIRLIN E	1	Range of Airline codes should be consistent with the IATA_CODE_AIRLINE in the Airline data variables. However have found that 4 values have LAX.	Use the Tail_number to determine which airline the LAX entered airlines are supposed to be. Use the website called FlightAware to manually find the correct airline (FlightAware, n.d.). All the values were actually Sky West Airlines coded as OO Unfortunately the missing value was also missing in FLIGHT_NUMBER, TAIL_NUMBER, ORIGIN_AIRPORT and DESTINATION_AIRPORT therefore this line was deleted.
FLIGHT_NUMBER	7	Four values have been entered correctly, look like actual tail numbers.	Ignore and will not remove as will not be included in any analysis.
TAIL_NUMBER	1	All the tail numbers should start with N, three have incorrect values	Ignore and will not remove as will not be included in any analysis.
ORIGIN_AIRPORT	6	This column includes departures from other airports that we are not interested in.	As LAX is the airport of interest in this analysis, missing data and data from other airports was deleted.
DESTINATION_ AIRPORT	42	There are 42 destinations with missing	Decided to keep and label as Missing, and only remove during analyses if irrelevant.

Variable	Missing values	Anomalies found	Cleaning approach
SCHEDULED DEPARTURE_TIME	3	Should be a 24 Hour clock time from 00:00 to 24:00 Sorted values an min and maximum within the 24 hour clock	These missing values will be deleted as we need these to determine departure delays
ACTUAL DEPARTURE_TIME	3	Should be a 24 Hour clock time from 00:00 to 24:00 Sorted values and found min and maximum within the 24 hour clock	These missing values will be deleted as we need these to determine departure delays
DEPARTURE_DELAY In Minutes	17	Need to check whether this has been calculated correctly (DEPARTURE_DELAY In Minutes = ACTUAL DEPARTURE_TIME - SCHEDULED DEPARTURE_TIME) Sorting data, it was found that 6 values entered incorrectly, in addition, one value = 700000 which doesn't seem correct.	Created a new column and named it DEPARTURE_DELAYMIN_CALC, used formula =CEILING((M2-L2)*1440,1), where L= ACTUAL DEPARTURE_TIME and K = SCHEDULED DEPARTURE_TIME, and converts the value into minutes. Any anomalies or missing data were corrected with the calculated value (particularly if there was a difference between actual value and calculated value that was -1<0<1), 35 values were replaced.
SCHEDULED ARRIVAL_TIME	0	Should be a 24 Hour clock time from 00:00 to 24:00 Sorted values and found min and maximum within the 24 hour clock	Nil
ACTUAL ARRIVAL_TIME	0	Should be a 24 Hour clock time from 00:00 to 24:00 Sorted values and found min and maximum within the 24 hour clock	Nil
ARRIVAL_DELAY In Minutes	11	There were a few anomalies. So calculated the value in two ways. See next column.	Created a new column and named it ARRIVAL_DELAYMIN_CALC, used formula = CEILING((P2-O2)*1440 ,1), where P= ACTUAL ARRIVAL_TIME and O = SCHEDULED ARRIVAL _TIME, and converts the value into minutes. Created a new column and named it ARRIVAL_DELAYMIN_CALC2, used formula to add AIR_SYSTEM_DELAY + SECURITY_DELAY + AIRLINE_DELAY + LATE_AIRCRAFT_DELAY WEATHER_DELAY Any anomalies or missing data were corrected with the calculated value (particularly if there was a difference between actual value and calculated value that was -1<0<1. 28 values were replaced.
CANCELLATIONS	24032	Minimal information regarding the values of this column, thus assuming there were no cancellations in 2015	Delete this column

Variable	Missing values	Anomalies found	Cleaning approach
AIR_SYSTEM_DELAY	0	Nil	For all these time points (AIR_SYSTEM_DELAY, SECURITY_DELAY, AIRLINE_DELAY, LATE_AIRCRAFT_DELAY, WEATHER_DELAY) there was no difference between the values ARRIVAL_DELAYMIN_CALC2 and ARRIVAL_DELAYMIN_CALC. So assumption is that these are all correct.
SECURITY_DELAY	0	Nil	Nil
AIRLINE_DELAY	0	Nil	Nil
LATE_AIRCRAFT_DELAY	0	Nil	Nil
WEATHER_DELAY	0	Nil	Nil

As outlined in Table 1, most missing values and anomalies were rectified. Specifically, missing DESTINATION_AIRPORT values were labelled as 'Missing' to return useful delay data for analysing causes due to time, day of the week and month. However, 'October' data was not available. No anomalies were found in the airline and airport data. Missing longitudes and latitudes in the airport data were not addressed as they were not needed for the initial analysis.

Exploring the data to confirm any outliers

The variables analysed for outliers were:

- DEPARTURE_DELAYMIN
- ARRIVAL_DELAYMIN
- AIR_SYSTEM_DELAY
- SECURITY_DELAY
- AIRLINE_DELAY
- LATE_AIRCRAFT_DELAY
- WEATHER_DELAY

Histograms, box plots and descriptive statistics were used to examine their distributions, identifying the centre, spread, shape and outliers (Section 3.2 in Appendix). The distributions between departure delay and arrival delay were very similar and thus only arrival delay will be used in the analysis. In addition, extreme values (departure delay of 1492 min and arrival delay of 1480 min) were found but retained, as 24-25 hour delays are not unusual for airlines. In addition, extreme delay values impact airline reputation and customer satisfaction and thus important. Refer also to Section 3.2 in Appendix for additional information.

1.4 Merge datasets

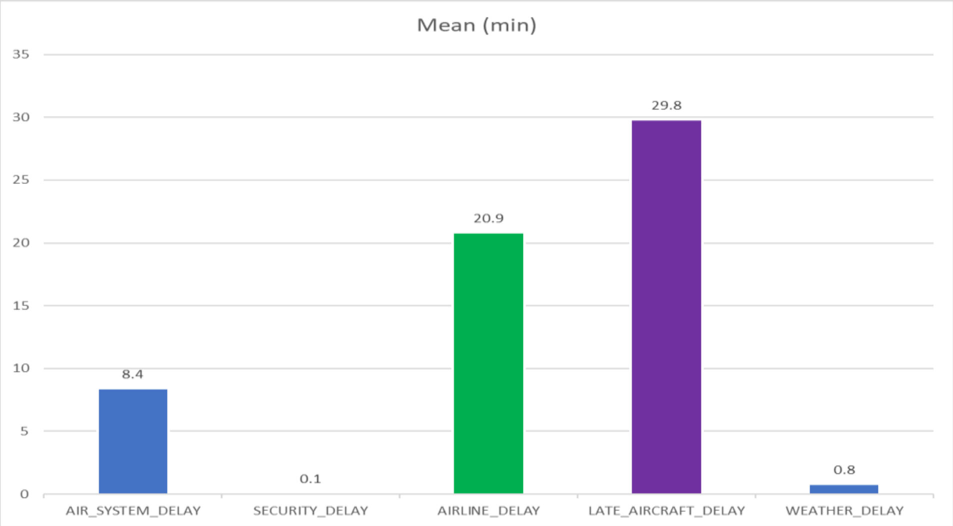
The four data tables were combined using SQLite within DB Browser. The steps involved are provided in Section 3.3 in Appendix.

1.5 Data analysis

Delay causes

The first step in determining arrival delay causes was comparing the mean delays for the year (Figure 1). This was more effective than using medians (Figure 24 in Appendix). The top two causes were late-aircraft and airline delay, followed by air-system delay. Weather and security delays were minimal overall.

Figure 1 Mean of delay causes



Airline & arrival delays

Given that airline, late aircraft and air-system delays were the highest causes, their distributions across airlines was determined (Figure 2 and Figure 3). Figure 3 includes the number of flights while Figure 4 shows the average flight arrival delays.

Figure 2 Proportion of causes vs airlines

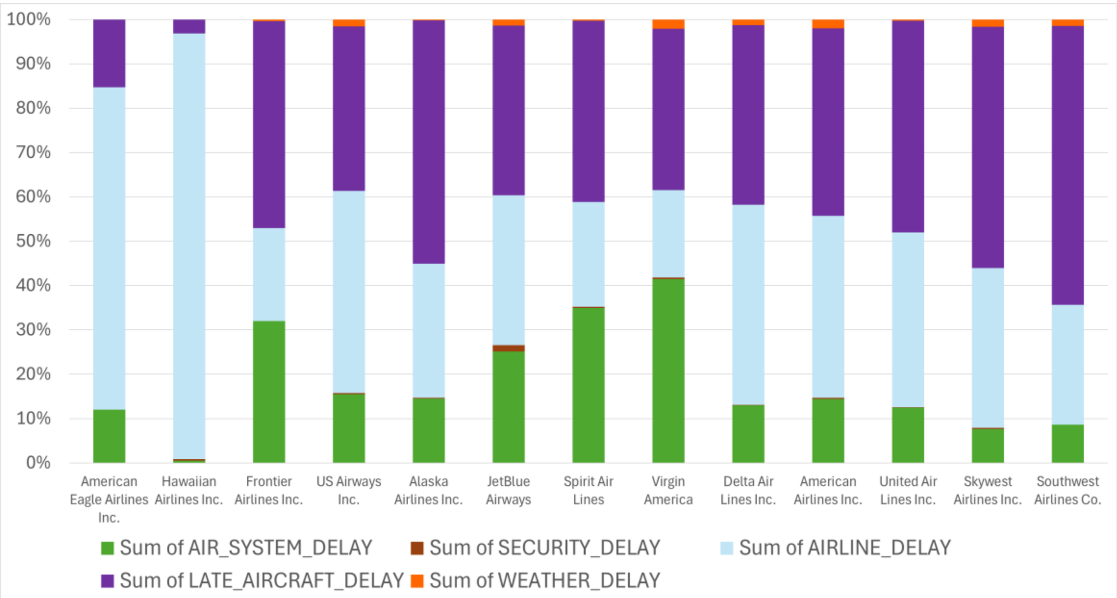


Figure 3 Sum of arrival delay causes vs airlines

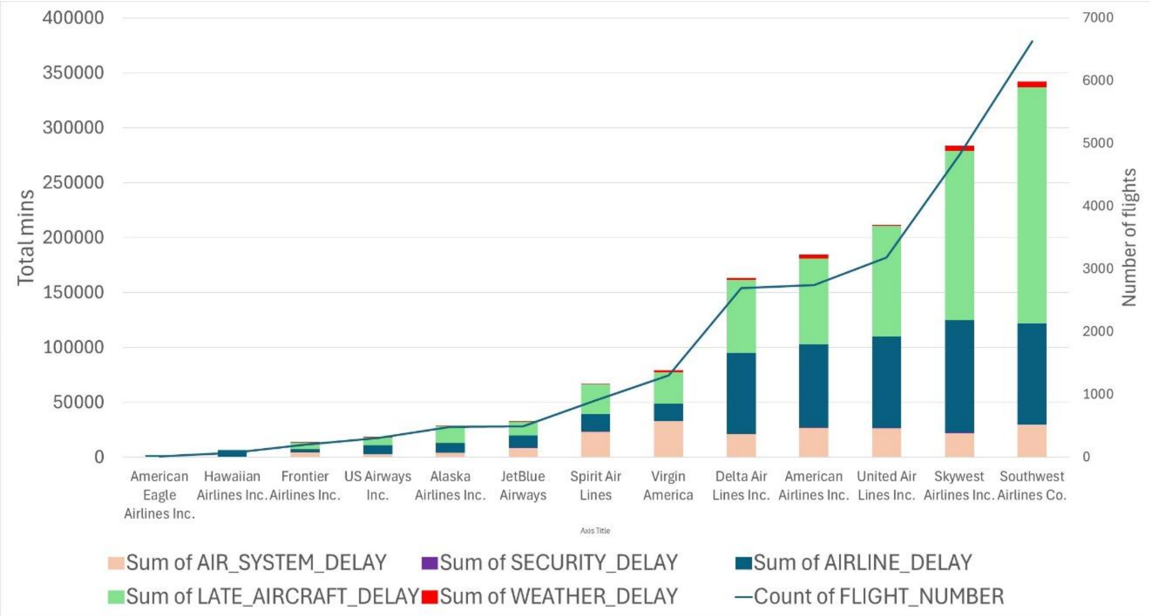


Figure 4 Average delay causes vs airlines

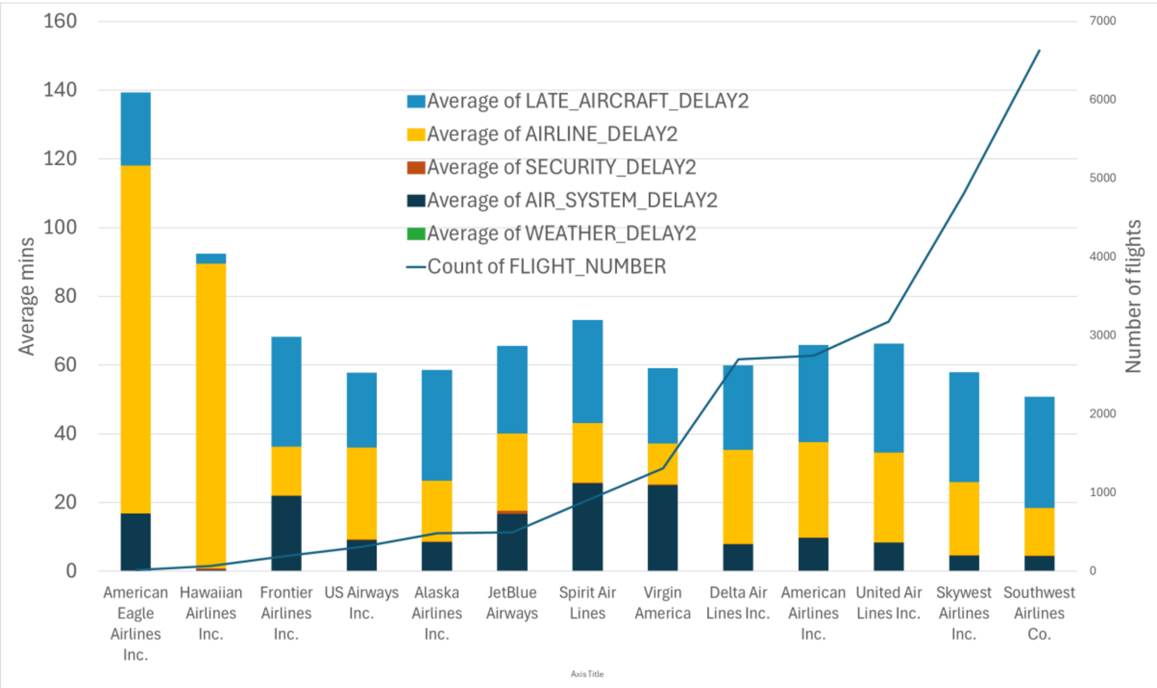
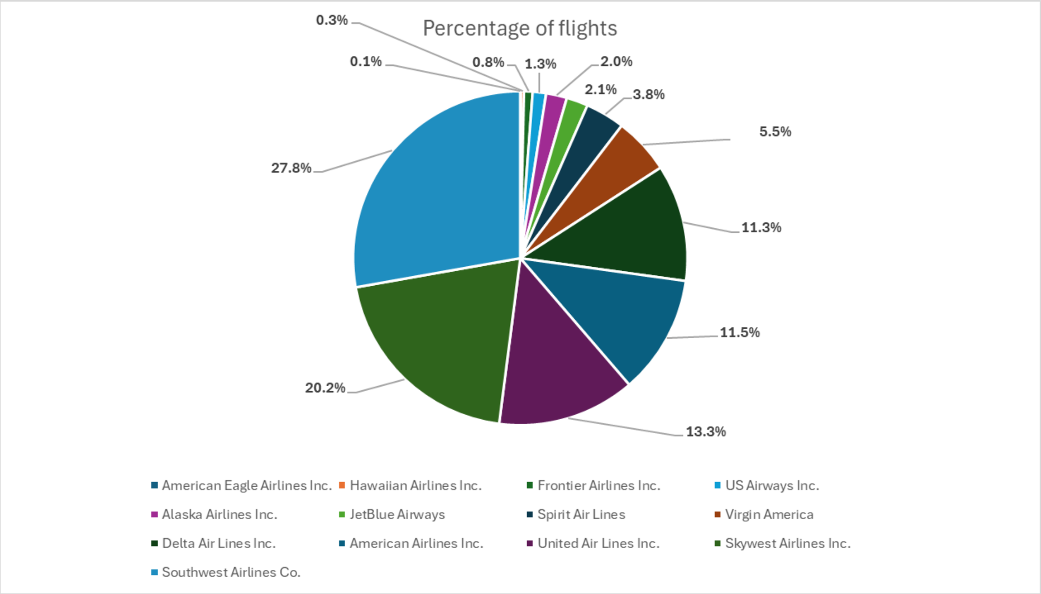


Figure 3 indicates a relationship between number of flights and total arrival delays (and arrival causes), with Southwest Airlines contributing the most delays. However, American Eagle and Hawaiian airlines have the highest average arrival delays despite having the fewest flights per year (Figure 4). For these airlines, the greatest contributing cause is airline delay. Skywest and Southwest airlines who provide the most flights (Figure 5), have the lowest average delays (Figure 4). With a greater proportion of the delays due late-aircraft delay.

Figure 5 Proportion of flights vs airlines



There was no relationship between average arrival delays and proportion of flights as shown in Figure 25 and Figure 26 in Appendix.

Scatter plots and Pearson's correlation coefficients (Figures 6-11) revealed strong positive relationships between the number of flights and total arrival delay, airline delay, and late-aircraft delay. Thus, the more flights an airline has the greater the arrival delay. This strength is mainly due to airline and late-aircraft delay. There was a moderate positive strength between flights and air-system delay and a weak positive relationship between flights and security delay. Surprisingly, a strong positive relationship was found between flights and weather delay, although weather events are beyond the airline's control. However, the overall average delay due to weather was very low and did not warrant further investigation.

Figure 6 Relationship between number of flights and total arrival delay

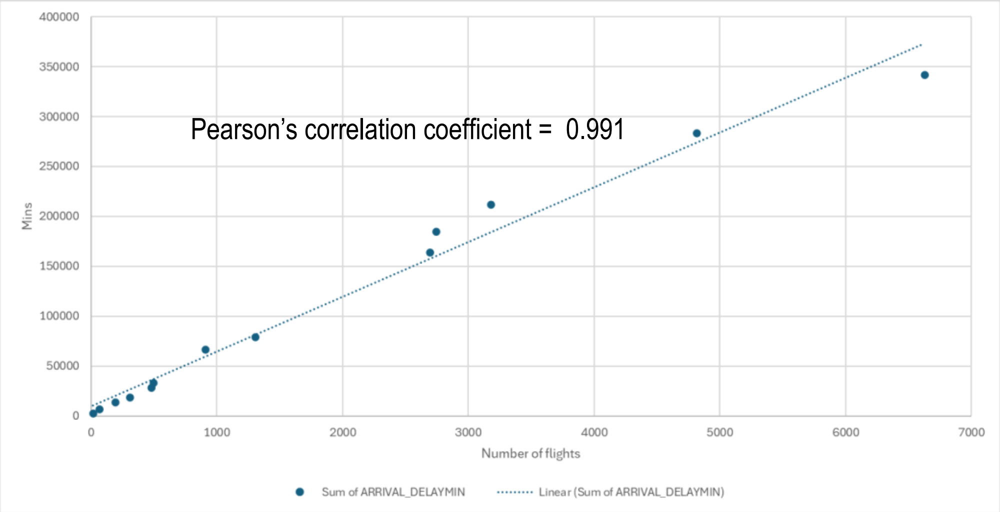


Figure 7 Relationship between number of flights and air system delay

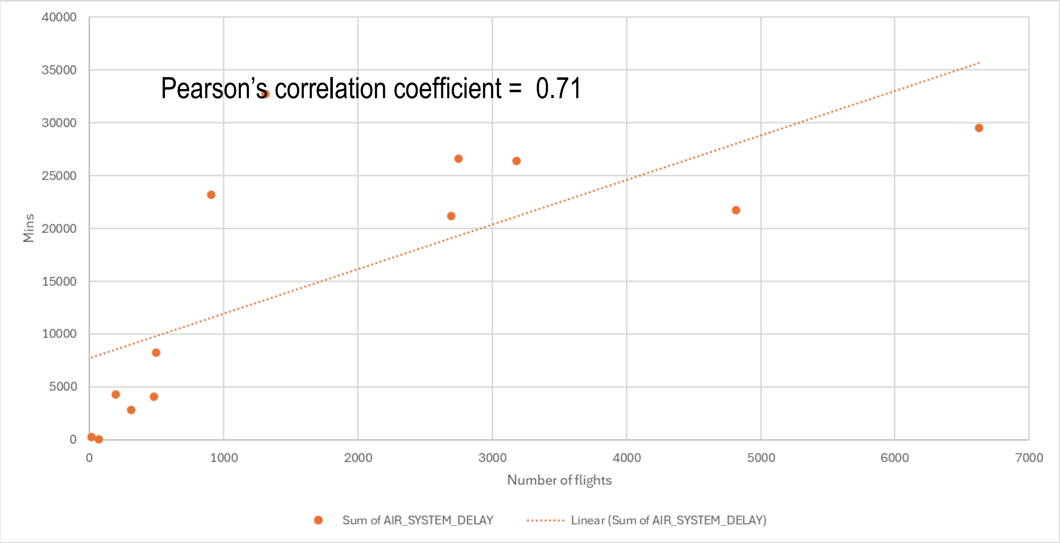


Figure 8 Relationship between number of flights and airline delay

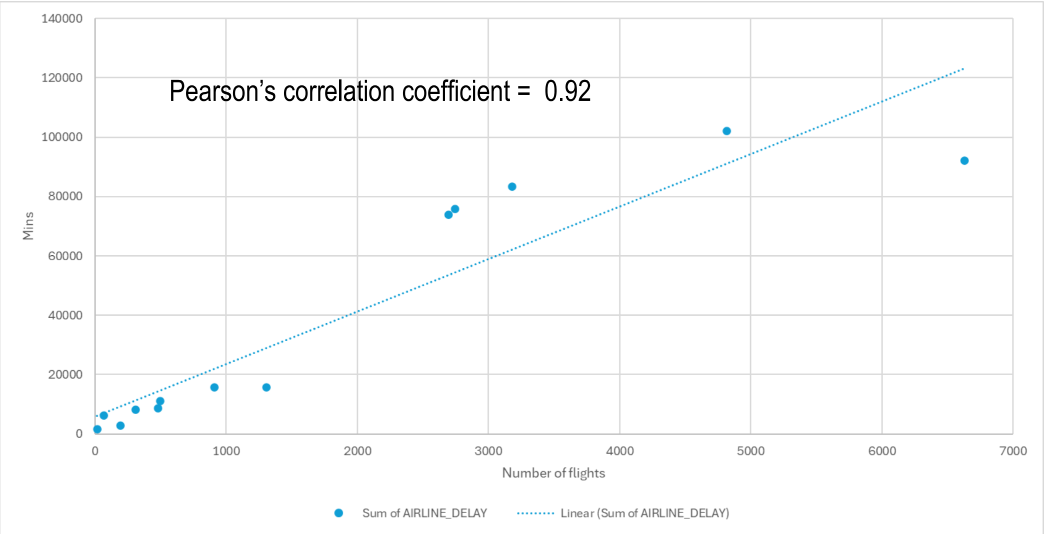


Figure 9 Relationship between number of flights and late-aircraft delay

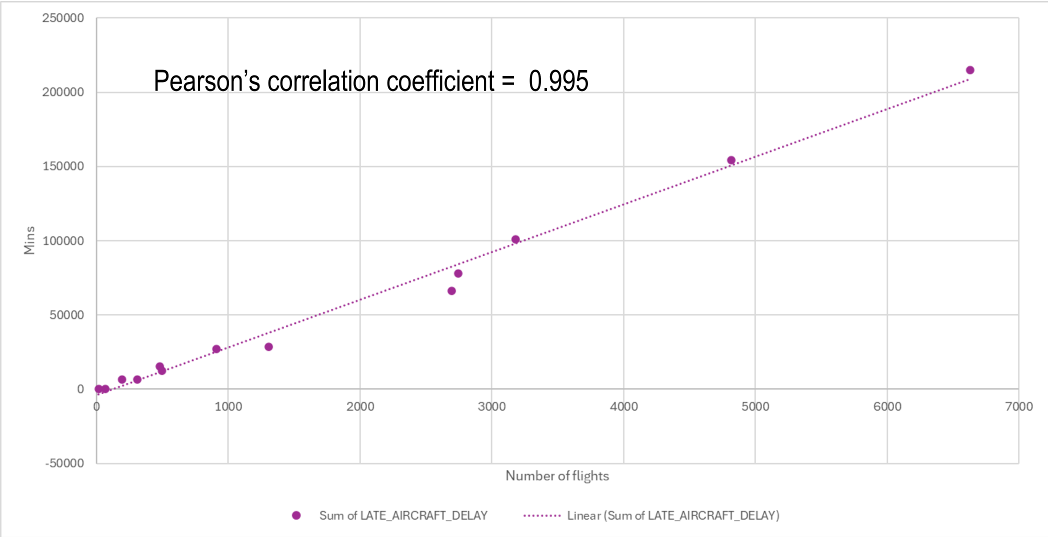


Figure 10 Relationship between number of flights and security delay

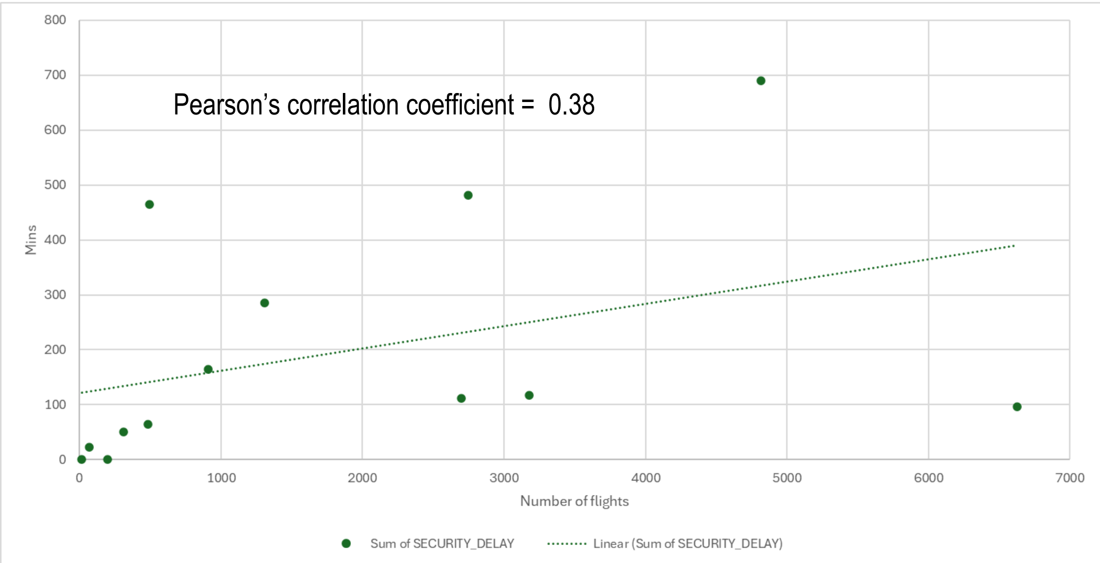
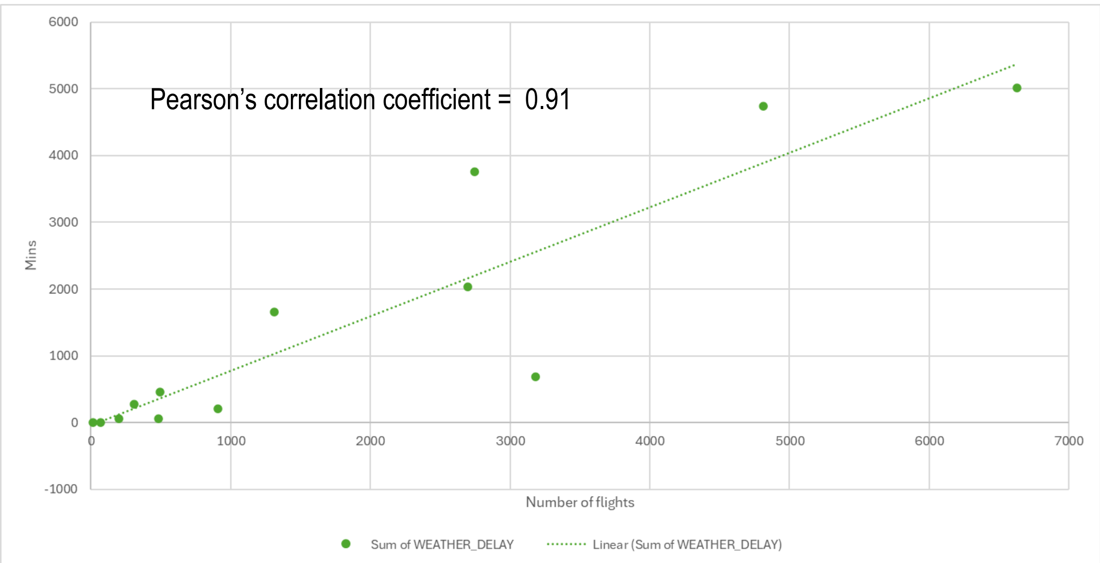


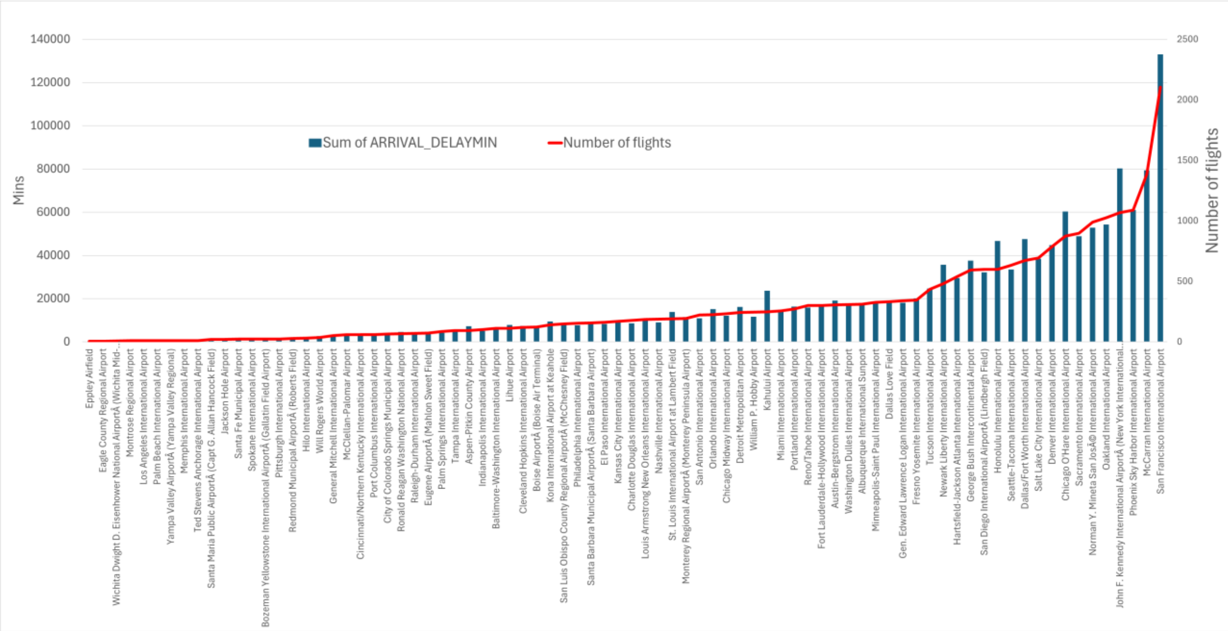
Figure 11 Relationship between number of flights and weather delay



Airports & arrival delays

The busiest airports were found to have the most arrival delays (Figure 12).

Figure 12 Total arrival delay vs airport



States & arrival delays

States with more airports experienced greater arrival delays, though the relationship was moderate (Figure 13). The top four states with the highest average delays were New York, Hawaii, Kansas and Alaska (Figure 14). Except Hawaii, these states have only one airport each. Delays in New York were mainly due to air-system issues, Hawaii faced late-aircraft and airline delay, Kansas mostly airline delays and Alaska experiences mostly late-aircraft delay.

Figure 13 Total arrival delay vs state

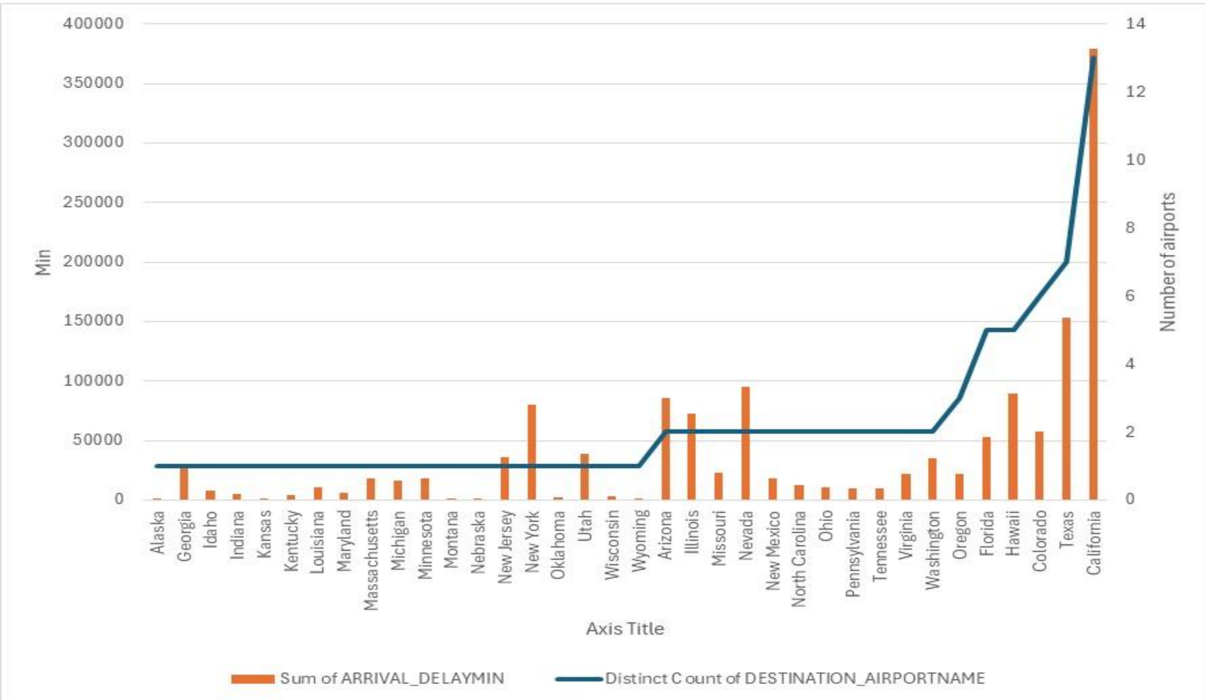
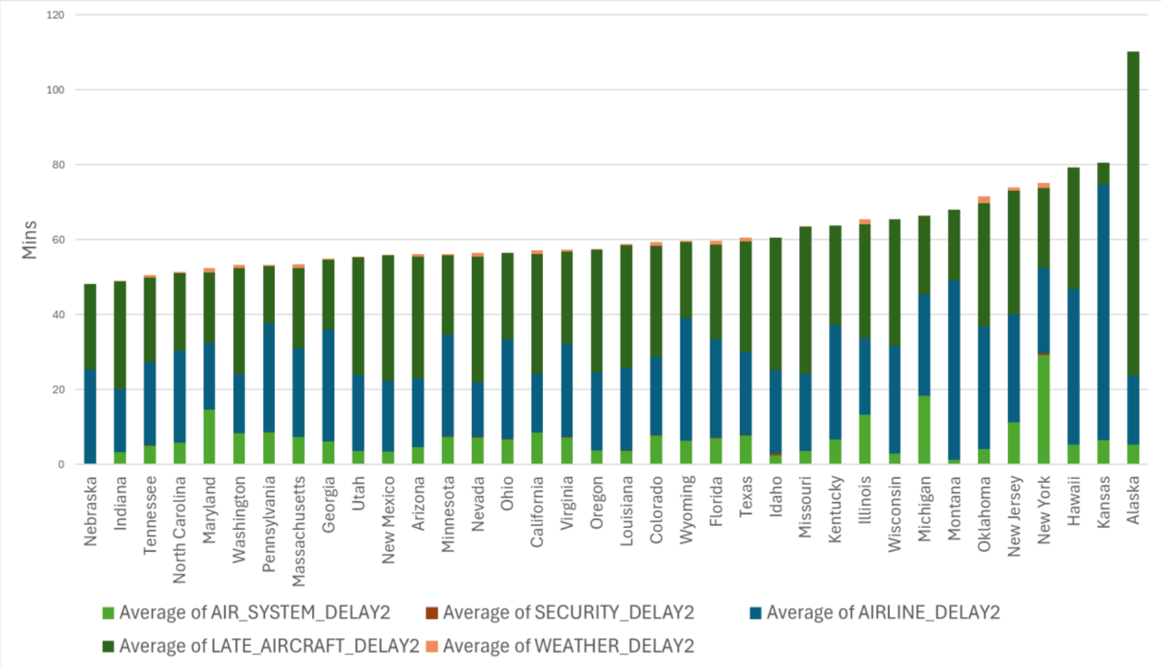


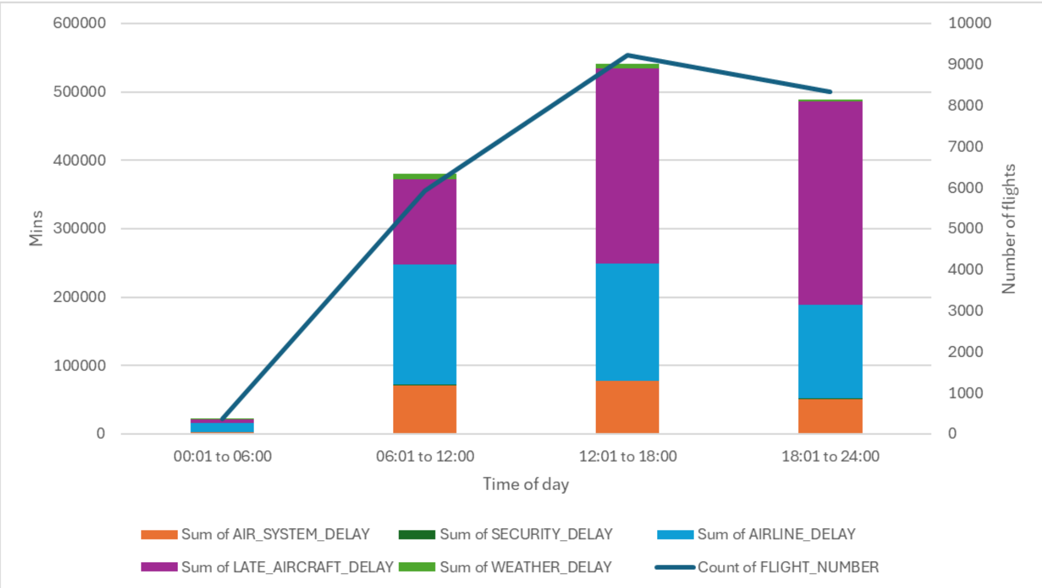
Figure 14 Average arrival delay vs state



Time of day & arrival delays

The busiest time of day was between 12 pm to 6pm which corresponded with the highest number of arrival delays.

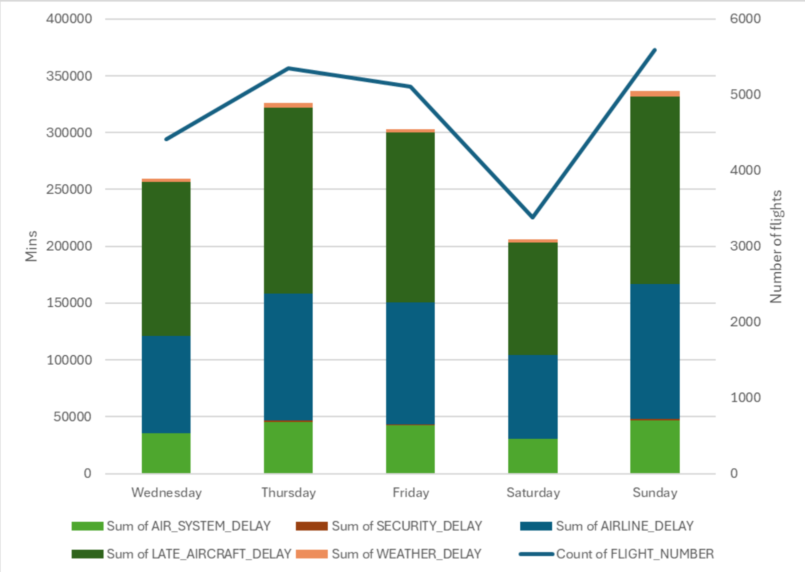
Figure 15 Total arrival delay vs time of day



Day of the week & arrival delays

The days with the most arrival delays were Thursday, Friday and Sunday corresponding to the number of flights on those days (Figure 16). However, daily average arrival delays were quite similar (Figure 27 in Appendix).

Figure 16 Total arrival delay vs day of the week



Month of the year & arrival delays

Monthly arrival delays also corresponded with the number of flights (Figure 17).

Figure 17 Total arrival delay vs month

1.6 Conclusion

The analysis revealed that the highest delay causes were late-aircraft (caused by previous flights arriving late) and airline delays (due to issues within the airline’s control, such as maintenance or crew problems), followed by air-system delays (related to non-extreme weather, airport operations, heavy traffic and air traffic control). To improve customer experience and service quality, airlines should focus on reducing these delays by optimising aircraft turnaround times and improving operational efficiency. Additionally, the busiest airports and states with the highest average delays, such as New York and Hawaii, should be targeted for infrastructure improvements and better resource allocation.

The busiest time of day, between 12 pm and 6 pm, corresponded with the highest number of arrival delays. To minimise bottlenecks, airlines could consider spreading out flight schedules more evenly throughout the day. Similarly, the days with the most arrival delays were Thursday, Friday and Sunday, which corresponded with higher flight numbers. Adjusting flight schedules on these days could also help reduce delays.

Monthly arrival delays also corresponded with the number of flights, suggesting that airlines should plan for peak travel times and allocate additional resources during these periods. By addressing these key factors, the company can enhance customer experience, improve service quality, reduce costs, and minimise landing and departure bottlenecks.

To avoid delays, passengers should consider flying earlier in the day and avoid peak travel days like Thursday, Friday and Sunday. Additionally, they should be prepared for potential delays when flying to busy airports or during peak travel months which correspond to the holiday seasons (January, July, December. Passengers should also consider flying Skywest and Southwest Airlines who have the lowest delays on average and avoid American eagle and Hawaiian airlines. In addition, they should plan ahead and expect delays when flying to New York, Hawaii, Kansas and Alaska.

Obtaining passenger demographics, flight crew schedules, real-time traffic data and customer feedback would provide further insights into causes of flight delays.

2 References

Bureau of Transportation Statistics (n.d.) *Understanding Reporting Causes of Flight Delays and Cancellations*. Available at: https://www.bts.gov/topics/airlines-and-airports/understanding-reporting-causes-flight-delays-and-cancellations (Accessed: 18 August 2024).

FlightAware (n.d.) *FlightAware - Flight Tracker*. Available at: https://www.flightaware.com/ (Accessed: 18 August 2024).

Ong, J. (n.d.) *List of US States*. Available at: https://github.com/jasonong/List-of-US-States/blob/master/states.csv (Accessed: 18 August 2024).

3 APPENDIX

3.1 Data provided

Flight data

File Name = Flightdata.xls
Number of columns = 21
Number of rows of data = 24032

Table 2 Variables of flight data file

Variable	Description	Type	Class of data	Notes
YEAR	Year	number	Categorical	
MONTH	Month	number	Categorical	
DAY	Day	number	Categorical	
DAY_OF_WEEK	Day of the week, 1=Monday, 2=Tuesday to 7=Sunday	number	Categorical ordinal	
IATA_CODE_AIRLINE	Airline code	string	Categorical	
FLIGHT_NUMBER	Airline flight number	number	Categorical	
TAIL_NUMBER	Airline tail number	string	Categorical	
ORIGIN_AIRPORT	Where the flight departed from	string	Categorical	We are only interested in flights departing from LAX
DESTINATION_AIRPORT	Where the flight landed	string	Continuous	IATA Codes in Airport data
SCHEDULED DEPARTURE_TIME	Planned departure	Date & Time	Continuous	Should be a 24 Hour clock time from 00:00 to 24:00
ACTUAL DEPARTURE_TIME	Actual departure	Date & Time	Continuous	Gate departure time is the instance when the pilot releases the aircraft parking brake after passengers have loaded and aircraft doors have been closed (https://www.transtats.bts.gov/glossary.asp) Should be a 24 Hour clock time from 00:00 to 24:00
DEPARTURE_DELAY In Minutes	Actual departure delay in min	number	Continuous	(DEPARTURE_DELAY In Minutes = ACTUAL DEPARTURE_TIME - SCHEDULED DEPARTURE_TIME) Relabelled as DEPARTURE_DELAYMIN
SCHEDULED ARRIVAL_TIME	Planned arrival	Date & Time	Continuous	

Variable	Description	Type	Class of data	Notes
ACTUAL ARRIVAL_TIME	Actual arrival	Date & Time	Continuous	Gate arrival time is the instance when the pilot sets the aircraft parking brake after arriving at the airport gate or passenger unloading area (https://www.transtats.bts.gov/glossary.asp)
ARRIVAL_DELAY In Minutes	Actual arrival delay in min (Should equal actual time – scheduled time)	number	Continuous	Need to check whether this has been calculated correctly (ARRIVAL _DELAY In Minutes = ACTUAL ARRIVAL _TIME - SCHEDULED ARRIVAL _TIME) ARRIVAL _DELAY In Minutes= AIR_SYSTEM_DELAY+ SECURITY_DELAY+ AIRLINE_DELAY+ LATE_AIRCRAFT_DELAY+ WEATHER_DELAY) Relabelled as ARRIVAL_DELAYMIN Arrival delay equals the difference of the actual arrival time minus the scheduled arrival time. A flight is considered on-time when it arrives less than 15 minutes after its published arrival time[1].
CANCELLATIONS	Whether the flight was cancelled	NA	NA	Minimal information regarding the values of this column, thus assuming there were no cancellations in 2015
AIR_SYSTEM_DELAY	Delay in min (assuming) due to air systems	number	Continuous	Delays and cancellations attributable to the national aviation system that refer to a broad set of conditions, such as non-extreme weather conditions, airport operations, heavy traffic volume, and air traffic control. [1]
SECURITY_DELAY	Delay in min (assuming) due to security	number	Continuous	Delays or cancellations caused by evacuation of a terminal or concourse, re-boarding of aircraft because of security breach, inoperative screening equipment and/or long lines in excess of 29 minutes at screening areas. [1]
AIRLINE_DELAY	Delay in min (assuming) due to administrative factors of the airline	number	Continuous	The cause of the cancellation or delay was due to circumstances within the airline's control (e.g. maintenance or crew problems, aircraft cleaning, baggage loading, fueling, etc.). [1]
LATE_AIRCRAFT_DELAY	Delay in min (assuming) due to aircraft while in flight	number	Continuous	A previous flight with same aircraft arrived late, causing the present flight to depart late. [1]
WEATHER_DELAY	Delay in min (assuming) due to weather	number	Continuous	Significant meteorological conditions (actual or forecasted) that, in the judgment of the carrier, delays or prevents the operation of a flight such as tornado, blizzard or hurricane. [1]

[1]United States Department of Transportation, Bureau of Transportation Statistics (2024), *Understanding the reporting of causes of flight delays and cancellations,* (*https://www.bts.gov/topics/airlines-and-airports/understanding-reporting-causes-flight-delays-and-cancellations*), *downloaded August 2024.*

Airlines data

File Name = airlines.xls
Number of columns = 2
Number of rows of data = 14

Table 3 Variables of airline data file

Variable	Description	Type	Class of data	Notes
IATA_CODE	International Air Transport Association's Airline code	number	Categorical	Use this key to link DESTINATION_AIRPORT in flightdata with the name of the airline Renamed as IATA_CODE_AIRLINE
AIRLINE	Name of airline	string	Categorical	

Airports data

File Name = airports.xls
Number of columns = 7
Number of rows of data = 322

Table 4 Variables of airport data file

Variable	Description	Type	Class of data	Notes
IATA_CODE	International Air Transport Association's Airport/Location code	number	Categorical	Use this key to link IATA_CODE_AIRPORT in the flightdata DESTINATION_AIRPORT
AIRPORT	Name of airport	string	Categorical	
CITY	City the airport is located	string	Categorical	
STATE	State the airport is located	string	Categorical	
COUNTRY	Country the airport is located	string	Categorical	
LATITUDE	Latitude map coordinates	number	Categorical	There are missing data however this variable not required for analysis.
LONGITUDE	Longitude map coordinates	number	Categorical	There are missing data however this variable not required for analysis.

USAstate data

File Name = USAstates.xls
Number of columns = 2
Number of rows of data = 51

Table 5 Variables of USA states data

Variable	Interpretation of variables	Type	Class of data	Notes
STATENAME	Name of State	string	Categorical	
STATE_CODE	Abbreviation	string	Categorical	Link to STATE in airport data file.

3.2 Assess quality of the data

Excel:

- Sorting each column (largest to smallest then smallest to largest for numerical and AtoZ then ZtoA for strings)
- Using Pivot tables to count the values if they were categorical

Python:

Checked missing data and uniqueness of values.

```
#check null values
flightdata.isnull().sum()
```

```
YEAR                          0
MONTH                         0
DAY                           0
DAY_OF_WEEK                   3
IATA_CODE_AIRLINE             1
FLIGHT_NUMBER                 7
TAIL_NUMBER                   1
ORIGIN_AIRPORT                6
DESTINATION_AIRPORT          42
SCHEDULED_DEPARTURE_TIME      3
ACTUAL_DEPARTURE_TIME         3
DEPARTURE_DELAY              17
SCHEDULED_ARRIVAL_TIME        0
ACTUAL_ARRIVAL_TIME           0
ARRIVAL_DELAY                11
CANCELLATIONS             24032
AIR_SYSTEM_DELAY              0
SECURITY_DELAY                0
AIRLINE_DELAY                 0
LATE_AIRCRAFT_DELAY           0
WEATHER_DELAY                 0
dtype: int64
```

```
#checked uniqueness of values for Year, Month, Day_of_week
columns = ['YEAR', 'MONTH', 'DAY', 'DAY_OF_WEEK', 'IATA_CODE_AIRLINE','ORIGIN_AIRPORT', 'DESTINATION_AIRPORT',]

for column in columns:
    print(f"{column}: {flightdata[column].nunique()}")
    print(f"{column}: {flightdata[column].unique()}")
```

```
YEAR: 3
YEAR: [2015 2025 2035]
MONTH: 14
MONTH: [ 1  2  3  4  5  6  7  8  9 11 12 15 18 30]
DAY: 31
DAY: [ 1  2  3  4  7  8  9 10 11 14 15 16 17 18 21 22 23 24 25 28 29 30 31  5
  6 12 13 19 20 26 27]
DAY_OF_WEEK: 8
DAY_OF_WEEK: ['12' '4' '5' '6' '7' '3' 'WEDS' '10' nan]
IATA_CODE_AIRLINE: 14
IATA_CODE_AIRLINE: ['AA' 'AS' 'B6' 'DL' 'F9' 'HA' 'NK' 'OO' 'UA' 'US' 'VX' 'WN' 'LAX' 'MQ'
 nan]
ORIGIN_AIRPORT: 9
ORIGIN_AIRPORT: ['LAX' 'DFW' 'IAH' 'ASE' 'SEA' 'SJC' 'SNA' 'BUR' 'ABQ' nan]
DESTINATION_AIRPORT: 80
DESTINATION_AIRPORT: ['DFW' 'BNA' 'MCO' 'CLT' 'MIA' 'STL' 'EGE' 'ORD' 'SEA' 'JFK' 'HNL' 'SLC'
 'ATL' 'DTW' 'OGG' 'MSP' 'SFO' 'LIH' 'OAK' 'IAH' 'SAN' 'SAT' 'SBA' 'RNO'
 'PHX' 'ASE' 'SMF' 'AUS' 'JAC' 'SJC' 'DEN' 'CLE' 'LAS' 'EWR' 'BOS' 'PHL'
 'DAL' 'HOU' 'TUS' 'DCA' 'PBI' 'PDX' 'MSY' 'ELP' 'MRY' 'BOI' 'IAD' 'MCI'
 'BWI' 'MDW' 'COS' 'SBP' 'CLD' 'LAX' 'MEM' 'RDU' 'BZN' 'KOA' nan 'FLL'
 'TPA' 'FAT' 'ABQ' 'PSP' 'MKE' 'EUG' 'IND' 'CMH' 'RDM' 'OMA' 'ANC' 'PIT'
 'CVG' 'SAF' 'OKC' 'ICT' 'SMX' 'MTJ' 'HDN' 'GEG' 'ITO']
```

Exploring the data to confirm any outliers

Departure and arrival delays have similar distributions (Figure 18, Figure 19, Figure 20 and Table 6).

Figure 18 Box-plot of departure and arrival delays

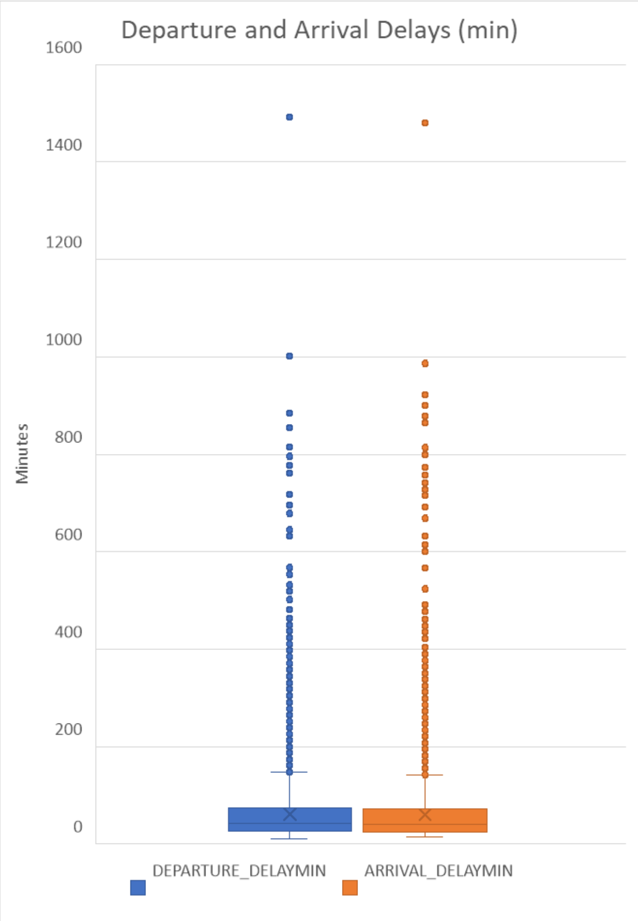


Figure 19 Histogram of departure delays

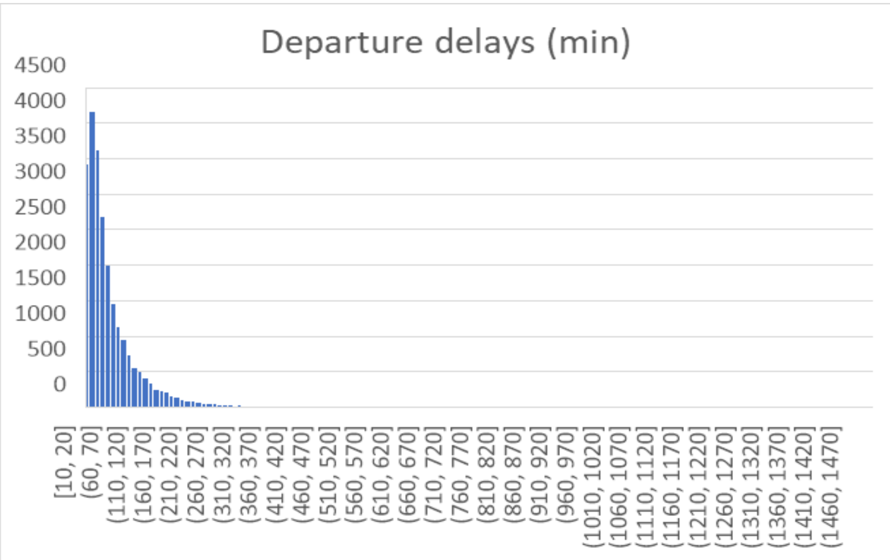


Figure 20 Histogram of arrival delays

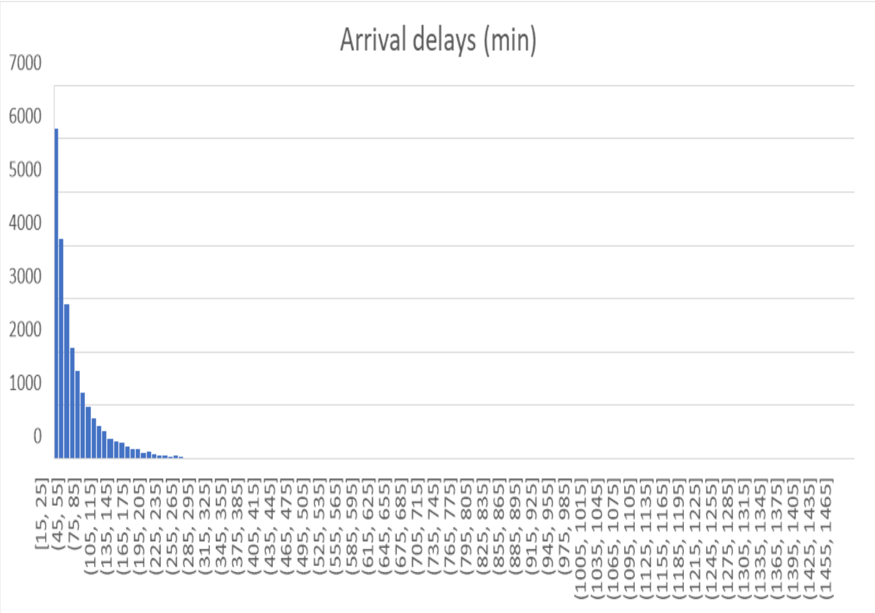


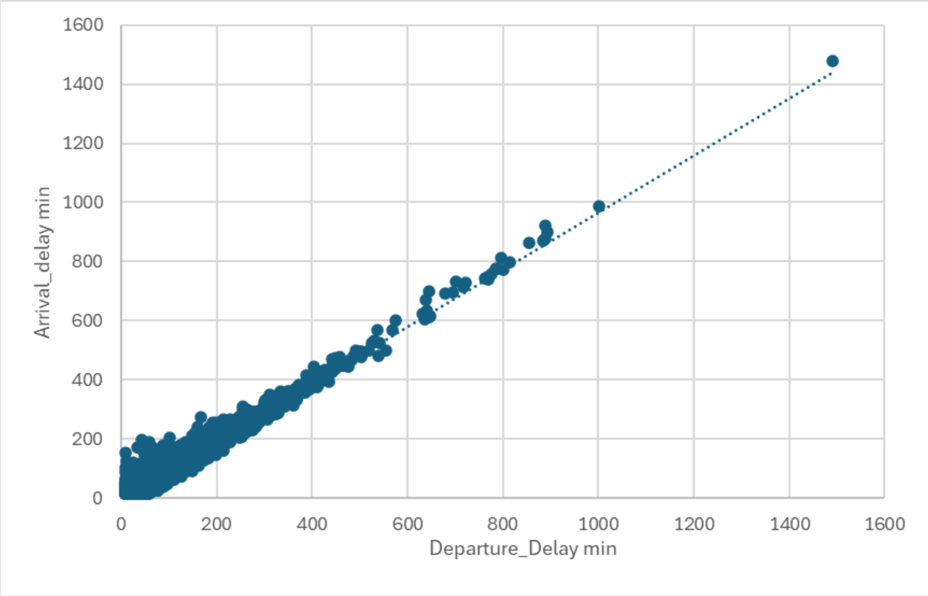
Table 6 Descriptive statistics for departure delay and arrival delay

	DEPARTURE_DELAYMIN	***ARRIVAL_DELAYMIN***
Mean	61.5	60.0
Median	43	41
Mode	25	17
Standard Deviation	60.7	59.9
Kurtosis	42.2	43.6
Skewness	4.4	4.5
Range	1482	1465
Minimum	10	15
Maximum	1492	1480
Confidence Level(95.0%)	0.77	0.76
Interquartile Range (IQR)	48	47

Both departure and arrival delay distributions were right-skewed due to outliers above the top whiskers (values outside 1.5 x IQR). High kurtosis (> 42), positive skewness (> 4), and means greater than medians confirmed this. An extreme value was found in both distributions (departure delay of 1492 min and arrival delay of 1480 min). Extreme values (departure delay of 1492 min and arrival delay of 1480 min) were found but retained, as 24-25 hour delays are not unusual for airlines. In addition, extreme delay values impact airline reputation and customer satisfaction and thus important.

The similarity in distributions was confirmed by a scatter plot (Figure 21) showing a strong positive relationship (Pearson correlation coefficient of 0.98). If we consider the causes of delay (refer to definitions in Table 2 in Appendix) they all affect the delay in the departure which in turn affects the arrival delay. As causes add up to the arrival delay, arrival delay will be used in the analysis.

Figure 21 Scatter-plot of arrival delay against departure delay



The distributions for the caused of delay (air_system_delay, security_delay, airline_delay, late_aircraft_delay and weather_delay) were also positively skewed with a high peak (Figure 22 and Table 7). What is noticeable in Figure 22, is that the extreme outlier is due to airline delay.

Figure 22 Box-plots of causes of delay

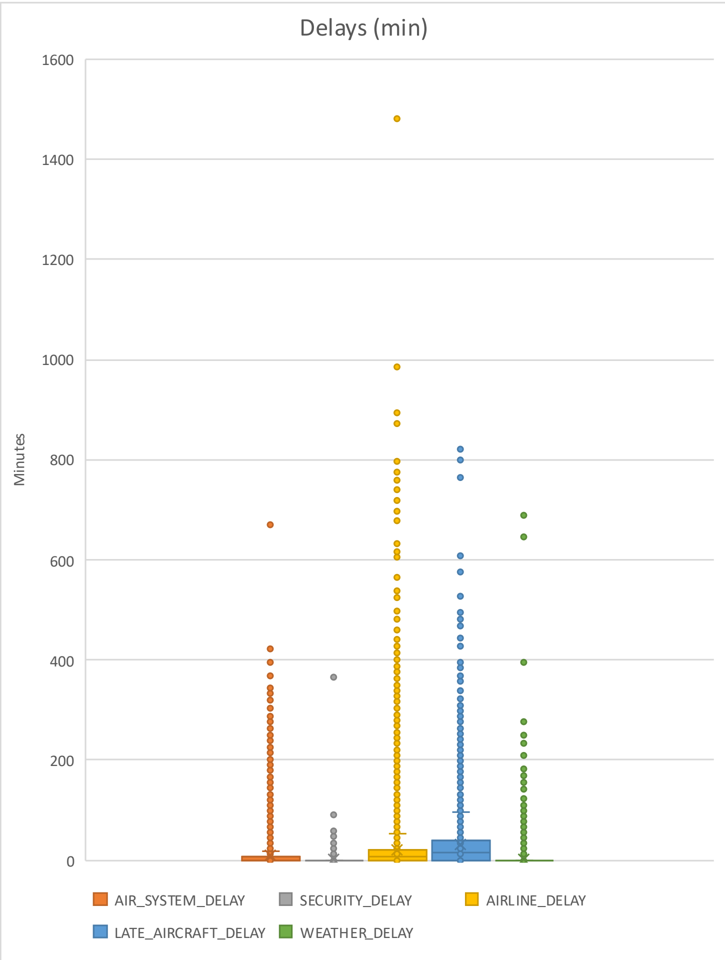


Table 7 Descriptive statistics for delay causes

	AIR_SYSTEM_ DELAY	SECURITY_ DELAY	AIRLINE_ DELAY	LATE_ AIRCRAFT_ DELAY	WEATHER_ DELAY
Mean	8.4	0.1	20.9	29.8	0.8
Median	0.0	0.0	7.0	16.0	0.0
Mode	0.0	0.0	0.0	0.0	0.0
Standard Deviation	24.0	3.1	45.9	45.2	10.3
Kurtosis	78.8	8047.7	117.4	25.2	1675.3
Skewness	7.0	75.5	7.8	3.6	32.6
Range	669.0	364.0	1480.0	820.0	689.0
Minimum	0.0	0.0	0.0	0.0	0.0
Maximum	669.0	364.0	1480.0	820.0	689.0
Confidence Level(95.0%)	0.3	0.0	0.6	0.6	0.1
Interquartile Range (IQR)	7.0	0.0	21.0	39.0	0.0

3.3 Merge datasets in SQLite

Relationship of tables

Flights and Airports (destination airport)

- Cardinality: A flight can have only one destination airport but a destination airport can have many flights
- Modality: A flight must have at least one destination airport but only one (I), but a destination airport can have no flights (O)

Flights and Airlines

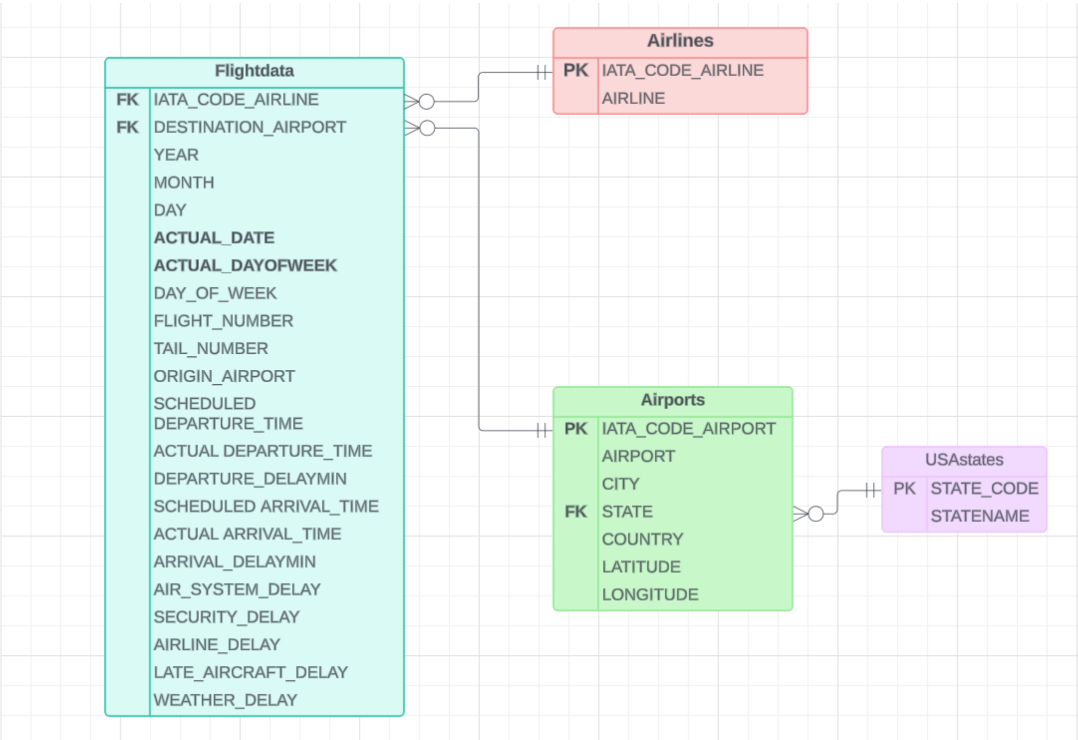
- Cardinality: A flight can have only one and only one airline but an airline can have many flights
- Modality: A flight must have at least one airline but only one (I), but an airline can have no flights (O)

Airport and State

- Cardinality: An airport is in one state but a state has many airports
- Modality: An airport can must be in one and only one state (I), and a state can have no airports (O)

The entity relationship diagram for the four datasets is provided in Figure 23, indicating the matching keys.

Figure 23 Entity Relationship Diagram for four datasets, Flightdata, Airlines, Airport and USAstates



Merge datasets

Prior to importing tables, flightdata.xls (which has been relabelled as flightdatalax.xls) was saved as an .csv file.

Refer to code below for importing and checking the import of the .csv files.

```sql
-- Import flightdatalax.csv, airlines.csv and airports.csv using 'file' menu import
-- check if imported correctly

select * from flightdatalax

```

	YEAR	MONTH	DAY	ACTUAL_DATE	ACTUAL_DAYOFWEEK	DAY_OF_WEEK	IATA_CODE_AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT
1	2015	3	20	20/03/2015	Friday	5	WN	491	7819A	LAX
2	2015	11	18	18/11/2015	Wednesday	3	WN	1751	7819A	LAX
3	2015	8	28	28/08/2015	Friday	5	WN	594	GRIFFIN	LAX
4	2015	6	14	14/06/2015	Sunday	7	AA	1529	N001AA	LAX
5	2015	1	29	29/01/2015	Thursday	4	AA	17	N002AA	LAX
6	2015	2	22	22/02/2015	Sunday	7	AA	1249	N002AA	LAX

```
Execution finished without errors.
Result: 23858 rows returned in 91ms
At line 1:
-- Import flightdatalax.csv, airlines.csv and airports.csv using 'file' menu import
-- check if imported correctly

select * from flightdatalax
```

```sql
select * from airports
```

	IATA_CODE_AIRPORT	AIRPORT	CITY	STATE	COUNTRY	LATITUDE
1	ADK	Adak Airport	Adak	AK	USA	51.87796
2	ADQ	Kodiak Airport	Kodiak	AK	USA	57.74997
3	AKN	King Salmon Airport	King Salmon	AK	USA	58.6768
4	ANC	Ted Stevens Anchorage International...	Anchorage	AK	USA	61.17432
5	BET	Bethel Airport	Bethel	AK	USA	60.77978
6	BRW	Wiley Post-Will Rogers Memorial ...	Barrow	AK	USA	71.28545

```
Execution finished without errors.
Result: 322 rows returned in 46ms
At line 5:
select * from airports
```

```sql
select * from airlines

```

	IATA_CODE_AIRLINE	AIRLINE
1	AA	American Airlines Inc.
2	AS	Alaska Airlines Inc.
3	B6	JetBlue Airways
4	DL	Delta Air Lines Inc.
5	EV	Atlantic Southeast Airlines
6	F9	Frontier Airlines Inc.
7	HA	Hawaiian Airlines Inc.

```
Execution finished without errors.
Result: 14 rows returned in 7ms
At line 6:
select * from airlines
```

```
7    select * from USAstates
8
```

	STATENAME	STATE_CODE
1	Alabama	AL
2	Alaska	AK
3	Arizona	AZ
4	Arkansas	AR
5	California	CA
6	Colorado	CO
7	Connecticut	CT

```
Execution finished without errors.
Result: 51 rows returned in 5ms
At line 7:
select * from USAstates
```

The datasets were combined using a left join to ensure all the flight data was kept in the merge. The SQL code is provided below. The resulting table called Allflightdata was exported and then saved as an .xls to conduct analyses.

```
10   -- Combine all the tables using an inner join and create a table
11   CREATE TABLE Allfightdata As
12   select * from flightdatalax f
13   LEFT JOIN airlines a on f.IATA_CODE_AIRLINE=a.IATA_CODE_AIRLINE
14   LEFT JOIN airports ap on f.DESTINATION_AIRPORT=ap.IATA_CODE_AIRPORT
15   LEFT JOIN USAstates u on ap.STATE=u.STATE_CODE
16
17
18
```

	YEAR	MONTH	DAY	ACTUAL_DATE	ACTUAL_DAYOFWEEK	DAY_OF_WEEK	IATA_CODE_AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED DEPARTURE_
1	2015	3	20	20/03/2015	Friday	5	WN	491	7819A	LAX	RNO	11:10
2	2015	11	18	18/11/2015	Wednesday	3	WN	1751	7819A	LAX	RNO	21:30
3	2015	8	28	28/08/2015	Friday	5	WN	594	GRIFFIN	LAX	BNA	18:55
4	2015	6	14	14/06/2015	Sunday	7	AA	1529	N001AA	LAX	ATL	15:35
5	2015	1	29	29/01/2015	Thursday	4	AA	17	N002AA	LAX	AUS	6:35
6	2015	2	22	22/02/2015	Sunday	7	AA	1249	N002AA	LAX	ATL	10:00

```
Execution finished without errors.
Result: 23858 rows returned in 573ms
At line 14:
select * from flightdatalax f
LEFT JOIN airlines a on f.IATA_CODE_AIRLINE=a.IATA_CODE_AIRLINE
LEFT JOIN airports ap on f.DESTINATION_AIRPORT=ap.IATA_CODE_AIRPORT
LEFT JOIN USAstates u on ap.STATE=u.STATE_CODE
```

3.4 Data analysis

Figure 24 Median of delay causes

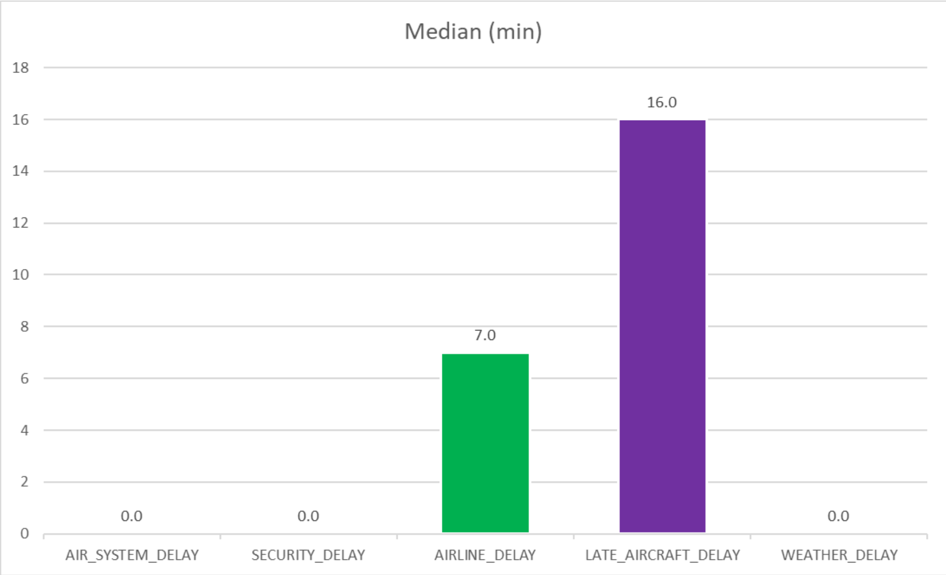


Figure 25 Average delays vs % of flights vs airline

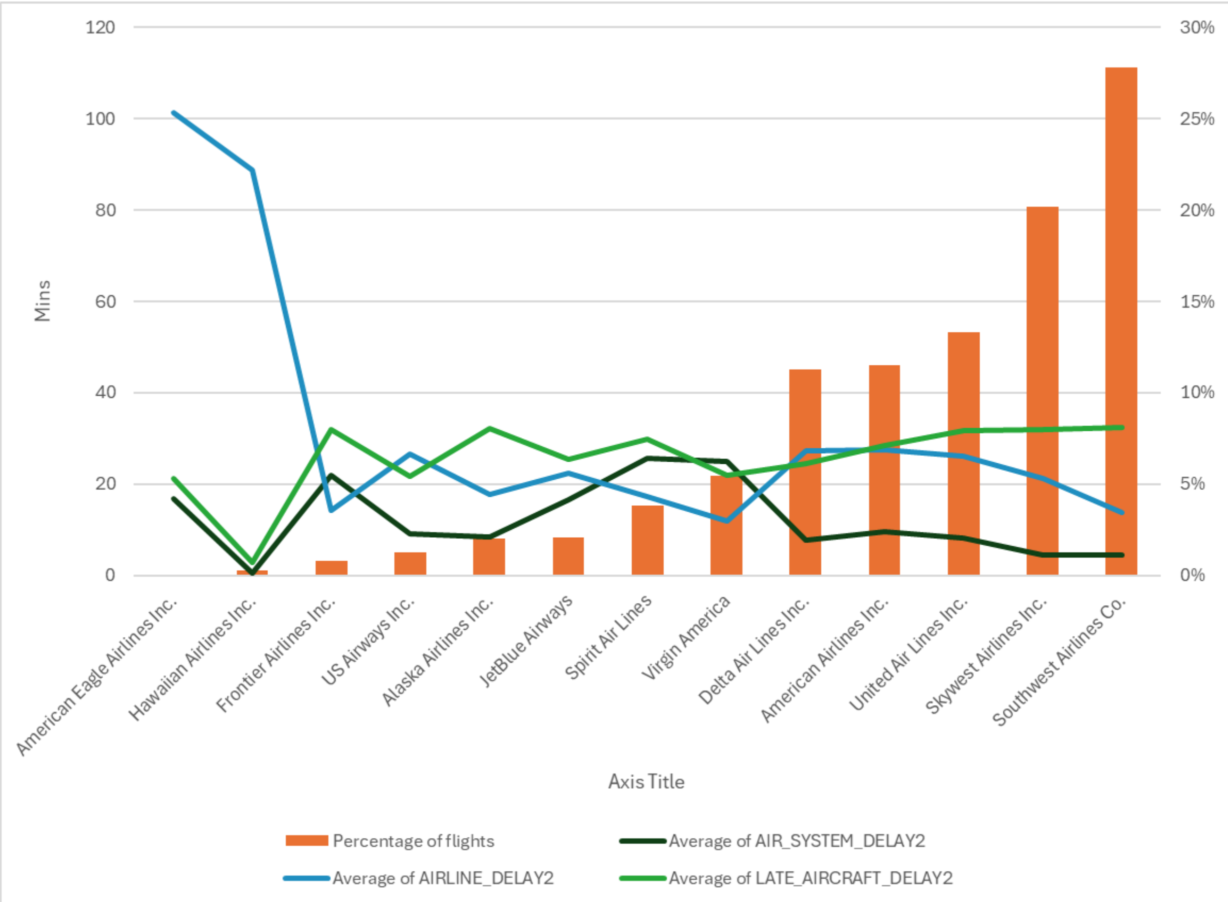


Figure 26 Average delays vs % of flights vs airline

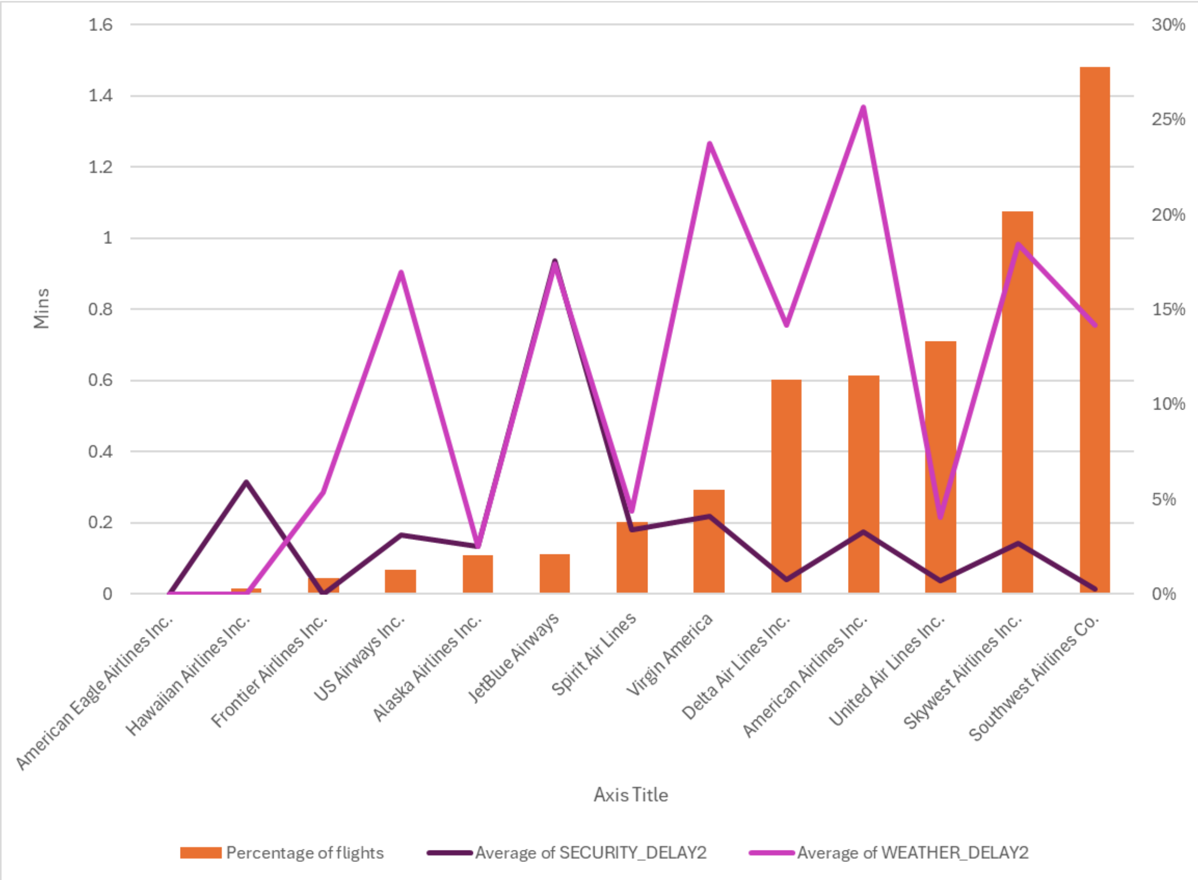


Figure 27 Average arrival delay vs day of the week